Assignment 3

BME.70034: Empirical Asset Pricing

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Instructor Information

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General Rule

- Submit two PDF files:
  - Architecture description (+ repository URL)
  - Results
- Your submission should be self-contained and fully executable — including any data downloads — producing the reported results.
  - Include 'requirements.txt' for the dependency information.
- Your code should include comments that explain the purpose and function of each chunk.
- You are encouraged to discuss this assignment with your classmates.
- You are more than encouraged to use LLM-assisted coding (i.e., vibe coding).
- If needed, obtain asset pricing factors from the Open Source Asset Pricing (https://www.openassetpricing.com/) or Ken French Data Library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).
- Make logical assumptions where necessary. For additional assumptions beyond what is provided, cite supporting academic papers.
- Copying and pasting another student's assignment is strictly forbidden.
- Read the syllabus and introduction slides carefully for more rules.

Sample

- U.S. common stock monthly returns from CRSP between 1971 and **2025**.
- NYSE stocks only.
- Follow Gu, Kelly, and Xiu (2020)
- If computational resources are limited:
  - First reduce the sample period (e.g., start from 2006).
  - If this is still insufficient, reduce the number of firms in the sample in a way that does not introduce bias in return predictability (e.g., random sampling, a fixed pre-selected subset, or proportional sampling across size and industry groups).

Goal and Outputs

- **Replicate** the following tables and figures of Gu, Kelly, and Xiu (2020).
  - **Table 1** "Monthly out-of-sample stock-level prediction performance"
    - Both the table and figure
    - Use only the following algorithms:
      - OLS+H, OLS-3+H, PCR, ENet+H, RF, NN2, NN4
    - Do only for "All" sample (skip Top 1000 and Bottom 1000) .
  - **Figure 4** "Variable importance by model"
    - Use only the following algorithms:
      - PCR, ENet+H, RF, NN2, NN4
  - **Figure 9** "Cumulative return of machine learning portfolios"
    - Use only the following algorithms:
      - OLS-3+H, PCR, ENet+H, RF, NN2, NN4
    - Also add SP500-R_f, as in the original figure and the shaded periods show NBER recession dates.
- Your final output PDF file should contain the two sets of above required outputs.
  - Set 1: using sample period 1971 to 2016
  - Set 2: using sample period 1971 to 2025

Instruction

1. Replicate the above tables and figures of Gu, Kelly, and Xiu (2020).
2. In addition to AI assistant, I encourage you to actively use/refer any of the following sources:
   - How to replicate Gu, Kelly, and Xiu, 2020 using R: Tidy Finance
   - Stock characteristic data (available up to 2021): Dacheng Xiu's Webpage

- SAS/Python Codes for the stock characteristic Data:
  - EquityCharacteristics
  - EquityCharacteristics Git
- Macroeconomic predictors: Tidy Finance